

Presuppositions and the Strongest Meaning Hypothesis

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Yeom, Jae-Il. 2001. **Presuppositions and the Strongest Meaning Hypothesis**, *Korean Journal of Linguistics*, 26-2, 331-362. In this paper, I claim that the Strongest Meaning Hypothesis explains why presuppositions tend to be projected. Presuppositions can be regarded as a means of conveying information, not as requirements for interpretation. When a sentence contains operators, presuppositions will have various scopes with respect to the operators depending on how far they are projected from their original positions. Then the interpretation of the sentence yields a set of readings, and the reading in which a presupposition is projected is stronger than the one in which it is not. In deriving a set of readings, I follow the principle of compositionality. This blocks some impossible readings which other theories may predict. Finally, I discuss disjunctive sentences, which show asymmetry in anaphora and symmetry in presupposition projection. (Hongik University)

Key words: presupposition, projection, strongest meaning hypothesis, accommodation, scope, informative, dynamic semantics

1. Introduction

Presuppositions tend to be projected unless they are canceled or suspended by contexts. Let's compare the following examples:

- (1) a. John does not regret leaving home.
b. John does not regret leaving home since he has never left home.

The first example presupposes that John has left home. The verb *regret* triggers the presupposition that John has left, and it is regarded as a fact unless the previous or following context tells/implies that it is not, or may not be, a fact. This is called **presupposition projection**. In the second example, the same presupposition is triggered, but it is not projected since it is overtly denied in the clause that follows. The two examples show that presuppositions tend to be projected unless they

are canceled by the context, as in the second example.

There are other examples that allow us to confirm this tendency:

- (2) a. It is possible that John regrets leaving home.
b. It is possible that John regrets leaving home, but it is also possible that he has never left home.
- (3) a. If John regrets leaving home, he will come back soon.
b. If John regrets leaving home, he will come back soon. But if he does not regret leaving home, he may still live with his family.

In (2a) and (3a), the presupposition that John has left home seems to project, so we believe that John has left home. On the other hand the same presupposition is suspended in (2b) and (3b). Note that when presuppositions are not canceled or suspended by the context, presuppositions project to the main context. In this paper, I will propose a theory of presupposition projection which can explain this tendency.

Before I start to discuss this, I would like to point out two more things about presuppositions. First, presuppositions cannot always be canceled.

- (4) ??John regrets leaving home, but he has not left home.

In (4) the triggered presupposition must project even though the second clause is not compatible with it. Our theory must be able to explain why it is so. Second, presuppositions may not be projected because they are preceded by some assertions which convey the same information.

- (5) John left home, and he regrets leaving home.

In some sense, the presupposition in (5) can be said to be "bound" by its antecedent. I will focus on cases where presuppositions are projected.

There are two main theories on presupposition projection: the Satisfaction Theory (Karttunen 1974, Heim 1983, Beaver 1992, 1995) and the Anaphora Analysis of presupposition projection (van der Sandt 1992, Zeevat 1993, Geurts 1995). In the Satisfaction Theory, a presupposition is projected by way of the repair of the local context for the satisfaction of the definedness condition. This theory basically requires local contexts to satisfy presuppositions, so the presuppositions do not have to be

projected beyond the local contexts. This is opposite to the general tendency of presupposition projection. Also what is the local context for the interpretation of an expression totally depends on actual formulation of interpretation rules, and unless the rules are independently motivated, they may be taken to be arbitrary, as pointed out by Soames (1989).

The Anaphora Analysis of presupposition projection assumes that presuppositions are anaphoric expressions. When they do not find their antecedents in the context, they establish anaphoric relations by being accommodated to the sites which are as far as possible from their original sites. Given that other anaphoric expressions tend to find their antecedents at the nearest possible sites, why do presuppositions show the opposite tendency? This theory also predicts that sites where their antecedents could be found should be sites for presupposition accommodation, but we will see that this is not always the case.

2. Presupposition Projection for Stronger Readings

2.1 The Strongest Meaning Hypothesis in Interpreting Reciprocals

In this section I will show that presupposition projection selects the strongest or a stronger reading of a sentence. To show that this is not peculiar to presupposition projection, I will discuss other phenomena. It has been claimed by several linguists (Dalrymple et al. 1994, Kim and Peters 1995, Krifka 1996, Winter 1996) that when there are several possible interpretations for an expression in which no structural ambiguities are involved, the strongest reading is selected as the meaning of the expression the speaker intends. The best example is the interpretation of reciprocals. Consider the following sentence and its interpretation:

- (6) The students saw each other.
 (7) $\exists ! X[students(X) \wedge \forall x[x \in X \rightarrow \forall y[y \in X \wedge y \neq x \rightarrow saw(x, y)]]]$

We cannot interpret sentence (8) in the same way, because with more than two plates stacked, it is impossible for one plate to be stacked on top of every other plate.

- (8) The plates are stacked on top of each other.
 (9) $?? \exists ! X[plates(X) \wedge \forall x[x \in X \rightarrow \forall y[y \in X \wedge y \neq x \rightarrow stacked_on_top_of(x, y)]]]$
 (9') $\exists ! X[plate(X) \wedge \forall x[x \in X \rightarrow \exists y[y \in X \wedge y \neq x \wedge stacked_on_top_of(x, y)]]]$

We consequently get a weaker interpretation, something like (9').

A similar observation has been made in the interpretation of plural predicates and donkey sentences. Here are examples of plural predicates. See Krifka (1996) for donkey sentences.

- (10) The ducks are swimming and quacking.
 (11) The ducks are swimming and flying.
 (12) $\exists ! X[ducks(X) \wedge \forall x[x \in X \rightarrow \forall P \in \{swim, quack\}[P(x)]]]$
 (13) $?? \exists ! X[ducks(X) \wedge \forall x[x \in X \rightarrow \forall P \in \{swim, fly\}[P(x)]]]$
 (13') $\exists ! X[ducks(X) \wedge \forall x[x \in X \rightarrow \exists P \in \{swim, fly\}[P(x)]]]$

Naturally, when a conjoined predicate applies to a sum individual, both conjuncts apply to all atoms, as in (10), which has the interpretation (12). However, if both predicates cannot apply to the same atom, as in (11), we cannot get a reading like (13), but a weaker interpretation (13'). Most importantly, when a stronger reading can be satisfied pragmatically, this is the reading which we get from the sentence. This is called the Strongest Meaning Hypothesis (= SMH hereafter). For an adequate analysis of reciprocals, the meaning of a reciprocal must be given not as one reading, but as a set of readings, and when a stronger one is not compatible with the context for pragmatic reasons, a weaker but the strongest one among the residual readings is selected.

In this paper, I am going to show that presupposition projection is another piece of evidence for the SMH. It has been observed that presuppositions tend to be projected unless they are incompatible with the context or already satisfied by intermediate contexts. First, I am going to show that the reading in which a presupposition is projected is stronger than the one in which it is not. Once this is shown, the preference of presupposition projection is naturally explained by resorting to the SMH. It is necessary to show that sentences with triggered presuppositions also carry a set of readings among which strength can be com-

pared. This leads us to the formal derivations of various readings induced by presuppositions. Finally it will be briefly discussed why a stronger reading is preferred from the general perspective of sentence processing.

2.2 Presupposition Projection and Stronger Readings

The notion of strength of a reading is based on the entailment relation: a stronger reading entails a weaker one, but not vice versa. I assume dynamic semantics within possible world semantics. The meaning of a sentence is regarded as a context change potential, or more specifically a partial function from sets of possible worlds to sets of possible worlds. The interpretation function is defined as follows:

- (14) A model $M = \langle W, F \rangle$, where W is a set of possible worlds, and F is a function from the set of sentences to the subsets of W
- (15) Let $s \subseteq W$ be an information state. Then
- (i) $s \parallel p \parallel = s \cap \{w \in W \mid w \in F(p)\}$ (p : an atomic sentence)
 - (ii) $s \parallel \phi \wedge \psi \parallel = s \parallel \phi \parallel \parallel \psi \parallel$
 - (iii) $s \parallel \neg \phi \parallel = s - s \parallel \phi \parallel$
 - (iv) $s \parallel \diamond \phi \parallel = \begin{cases} s & \text{if } s \parallel \phi \parallel \neq \emptyset \\ \emptyset & \text{if } s \parallel \phi \parallel = \emptyset \end{cases}$
 - (v) $s \parallel \phi \rightarrow \psi \parallel = s \parallel \neg(\phi \wedge \neg \psi) \parallel$

The interpretation of atomic sentence p takes an information state s as an argument, and yields a new information state which includes only possible worlds in which p is true, excluding all other possible worlds in s . When two sentences are conjoined, the first conjunct updates the information first, and then the result is updated with the second. In order to update an information state with a sentence $\neg \phi$, first the state is updated with the sentence ϕ , and then the resulting set of possible worlds are subtracted from the original information state. Veltman (1991) gives the interpretation rule (15iv) for the possibility operator. The original information state is first updated with the sentence ignoring the possibility operator. If the resulting information state becomes the empty set, the whole update leads to the empty set. Otherwise, the original information state remains the same. Finally, an implication

sentence is paraphrased with the negation and conjunction operators.

Now the entailment relation can be defined as follows:

(16) ϕ entails ψ iff

for every possible context $s \in \mathcal{S}(W)$, $s \Vdash \phi \Vdash s \Vdash \psi$.¹

The entailment relation must hold regardless of any particular context, and thus if we want to show that one reading entails another, we just have to show that for an arbitrary context, the former leads to a set of possible worlds that is at least as small as the one the latter does.

Now consider various cases in which presuppositions yield various readings. First, I will consider sentence (1a) again. The two readings of the sentence can be translated into propositional logic, as follows:

- (17) (i) $P_\phi \wedge \neg \phi$ ϕ : John regrets leaving home.
 (ii) $\neg(P_\phi \wedge \phi)$ P_ϕ : John left home.

The two propositions, P_ϕ and ϕ , divide worlds into four subsets according to their truth values in those worlds.

(18)

W	$W1$	$W2$	$W3$	$W4$
ϕ	0	0	1	1
P_ϕ	1	0	1	0

Since it is impossible that John did not leave home and he regrets leaving home, $W4$ must be excluded from the set of possible worlds. Let's assume that the current information state s is represented as $W1 \cup W2 \cup W3$. When P_ϕ is projected, s is updated with P_ϕ to $W1 \cup W3$, which is again updated to $W1$ with the assertion $\neg \phi$. When the presupposition is not projected, the information state s is updated by $\neg(P_\phi \wedge \phi)$ to $W1 \cup W2$, which is a superset of the result we get from the projected reading. Note that this holds regardless of any particular information state. In other words, it holds when the information state is $W1 \cup W2$.²

¹ Normally, the entailment relation is defined between sentences, but in a later discussion, it is also used for semantic entities. I believe this will not be confusing.

One extreme case is when the information state is $W1$ or its subset. In this case, both readings lead to the same updated information state, and a subset relation still holds. We do not have to consider the other cases because the information state must include some possible worlds in $W1$. Otherwise, the assertion and the presupposition would become the same proposition. Since the subset relation holds regardless of any particular information state, the projected reading entails the reading with the presupposition not projected.

This result can be easily predicted when we look at the two readings again. The possible worlds in $W1$ are excluded because P_ϕ necessarily holds when ϕ holds. Thus whenever we interpret ϕ , it has the same effect as interpreting $P_\phi \wedge \phi$. Therefore the two readings are as follows:

- (17') (i) $P_\phi \wedge \neg(P_\phi \wedge \phi)$
 (ii) $\neg(P_\phi \wedge \phi)$

Only the first reading additionally contributes the information conveyed by the presupposition outside the scope of the negation operator. When the domain outside the negation operator is monotone-increasing, the first reading becomes stronger. This is what we have observed above.

Next consider the possibility operator, assuming that the current information state s is $W1 \cup W2 \cup W3$, as given in (18). Considering the exclusion of the worlds in $W1$, sentence (2a) has the following readings:

- (19) (i) $P_\phi \wedge \circ(P_\phi \wedge \phi)$
 (ii) $\circ(P_\phi \wedge \phi)$

The first reading includes the additional information from the presupposition outside the scope of the possibility operator. If the scope outside the possibility operator is monotone-increasing, the first reading is stronger than the second. This is confirmed by the actual updates. In the first reading, the presupposition P_ϕ updates the initial information

2 When the information state is (a subset of) $W1 \cap W2$ or $W1$, the statement itself is not felicitous for pragmatic reasons. The assertion is not informative at all. I will ignore pragmatic factors for the time being.

state s to $W_1 \cup W_3$. The resulting information state may change to itself ($W_1 \cup W_3$) or to the empty set. On the other hand, the second reading may change s to itself ($W_1 \cup W_2 \cup W_3$) or to the empty set. When both readings lead to the empty set, the subset relation holds trivially. When both possibilities do not lead to the empty set, the first reading leads to a smaller set of possible worlds than the second one. I need to show that when the second reading results in the empty set, the first does too. The second reading results in the empty set if the information state is (a subset of) $W_1 \cup W_2$. In the same information state, the first reading also yields the empty set. Therefore the first reading always results in a smaller set of possible worlds than the second.

So far I have only dealt with cases where a sentence contains only one operator. Look at an example where there is more than one operator:

(20) It is possible that John does not regret leaving home.

- (21) (i) $P_\phi \wedge \diamond \neg(P_\phi \wedge \phi)$
 (ii) $\diamond (P_\phi \wedge \neg(P_\phi \wedge \phi))$
 (iii) $\diamond \neg(P_\phi \wedge \phi)$

Sentence (20) includes two operators, so three readings are obtained as given in (21). In (21i) the presupposition is added outside the scope of the possibility operator, and in (21ii) it is added within the scope of the same operator. Compared with (21iii), the strengths of (21i-ii) depend on the monotonicity of the domains where the presupposition is added.

Let's look at the informational property of the possibility operator. The possibility operator may be considered to be monotone-increasing:

- (22) a. It is possible that John had lunch.
 b. It is possible that John had lunch at noon.

Sentence (22a) is entailed by (22b), where more information is added within the scope of the possibility operator. This shows that the possibility operator introduces a monotone-increasing domain. Then we can expect that (21ii) is stronger than (21iii). Then which is stronger, (21i) or (21ii)? The possibility operator is regarded as a test. So an information state is tested with some information and if it is compatible with the information, it remains as it is. If it is not, it becomes the empty

set. Thus, insofar as the information is compatible with the information state, it does not actually change the information state. On the other hand, if the same information is added over the scope of the possibility operator, it does change the information state unless it is already contained in the context. Therefore (21i) is stronger than (21ii).

Let's see if this is confirmed by the actual example. When the initial information state is (a subset of) $W3$ in (18), the three readings (21i-iii) leads to the empty set. When the initial state is (a subset of) $W2 \cup W3$, but not a subset of $W3$, the readings (21i-ii) change it to the empty set while the reading (21iii) does not. When the initial state is (a subset of) $W1 \cup W2 \cup W3$, but not a subset of $W2 \cup W3$, the reading (21i) updates it to $W1 \cup W3$ while the other two readings do not change it at all. These are summarized in the following table:

(23)

initial state	$W3$	$W2 \cup W3$	$W1 \cup W2 \cup W3$
(21i)	$\emptyset$	$\emptyset$	$W1 \cup W3$
(21ii)	$\emptyset$	$\emptyset$	$W1 \cup W2 \cup W3$
(21iii)	$\emptyset$	$W2 \cup W3$	$W1 \cup W2 \cup W3$

This table might seem to show the entailment relations we expect. However, the initial information state must include some possible worlds in $W1$. Otherwise the presupposition and the assertion would coincide. With the information state including the possible worlds in $W1$, (21i-iii) lead to the same set of possible worlds. It may be the reason why there is no preference between the two readings.

When more than one presupposition is in a sentence, it will generate an even more complex entailment relation. Before considering a case where more than one presupposition interacts with (an) operator(s), let's look at a case where one presupposition is embedded in another.

(24) John regrets leaving his wife.

(25) ϕ = John regrets leaving his wife.

$P1_\phi$ = John left his wife

$P2_\phi$ = John has a wife

	W1	W2	W3	W4	W5	W6	W7	W8
ϕ	0	0	0	0	1	1	1	1
$P1_\phi$	0	0	1	1	0	0	1	1
$P2_\phi$	0	1	0	1	0	1	0	1

The three propositions divide the worlds into eight groups according to their truth values. However, $W3$ and $W7$ are excluded from the set of "possible" worlds since it is impossible to leave one's wife without having a wife. Similarly, $W5$ and $W6$ are excluded since one cannot regret leaving one's wife without leaving her. Considering the remaining possible worlds, $W1 \cup W2 \cup W4 \cup W8$, if $P1_\phi$ is true, $P2_\phi$ is too. Thus, if the former is projected, the latter is too. This holds regardless of particular presuppositions, and thus, we can generalize that when a presupposition is projected its embedded presuppositions are too.

When one presupposition is not embedded in another, each of them will yield different readings depending on their own projections. A conditional sentence is a good example for the case in point.

(26) If John's sister has grandchildren, his children are grown-ups.

This sentence consists of four propositions, which divide possible worlds into nine groups.³ We assume that the initial information state includes all possible worlds in the table below.

	W1	W2	W3	W4	W5	W6	W7	W8	W9
ϕ	0	0	0	0	0	0	1	1	1
P_ϕ	0	0	0	1	1	1	1	1	1
ψ	0	0	1	0	0	1	0	0	1
P_ψ	0	1	1	0	1	1	0	1	1

ϕ = John's sister has grandchildren

P_ϕ = John has a sister

ψ = John's children are grown-ups

P_ψ = John has children

3 Four propositions are expected to divide possible worlds into 16 groups, but 7 groups are excluded because they are "impossible" worlds.

From (26), we can derive six readings depending on whether the two presuppositions are projected or not. We can express them in terms of the four propositions, as in (27).

- (27) (i) $P_\phi \wedge P_\psi \wedge ((P_\phi \wedge \phi) \rightarrow (P_\psi \wedge \psi))$
 (ii) $P_\phi \wedge ((P_\phi \wedge \phi) \rightarrow (P_\psi \wedge \psi))$
 (iii) $P_\phi \wedge ((P_\phi \wedge \phi \wedge P_\psi) \rightarrow (P_\psi \wedge \psi))$
 (iv) $P_\phi \wedge ((P_\phi \wedge \phi) \rightarrow (P_\psi \wedge \psi))$
 (v) $((P_\phi \wedge \phi) \rightarrow (P_\psi \wedge \psi))$
 (vi) $((P_\phi \wedge P_\psi \wedge \phi) \rightarrow (P_\psi \wedge \psi))$

(27i) is the reading in which the two presuppositions are projected and (27v) is the one in which both stay in their original positions.

A conditional sentence can be paraphrased with the negation and the conjunction operators, as given in (15v). With this rule we can easily derive new information states from the six readings. I will leave this to the reader, and just give the results in the following table:

(28)

(27i)	(27ii)	(27iii)	(27iv)	(27v)	(27vi)
569	4569	45679	23569	1234569	12345679

Sentence (26) normally has the strongest reading (27i) unless that reading is canceled or neutralized in the context.

In (27) it is easy to see that reading (i) is the strongest, and that (ii) and (iv) are stronger than (v), because the presupposition is added to the main context. A question is whether readings (iii) and (vi) are stronger than (ii) and (v) respectively. According to (15v), a conditional is interpreted as a material implication. In this case, the antecedent clause is monotone-decreasing. Information in the antecedent clause only has the role of reducing the set of possible worlds which will be excluded from the context if the consequent clause does not hold in them. Thus, when the presupposition P_ψ is added to the antecedent clause, all possible worlds in which the presupposition does not hold remain in the resulting information state.

Stalnaker (1968) and Lewis (1973), however, claim that in natural

language the antecedent clause of a conditional is not monotonic.⁴ The two sentences in (29) do not have any entailment relationship.

- (29) a. If John comes to the party, the party will be a success.
 b. If John comes to the party late, the party will be a success.

Stalnaker and Lewis analyze conditionals as follows:

- (30) (i) A conditional 'if ϕ , ψ ' is true in a possible world w iff ψ is true in all ϕ -worlds maximally similar to w .
 (ii) A world w' is a ϕ -world maximally similar to w iff ϕ is true in w' and w' resembles w no less than any other world where ϕ is true.

In (29), when the party is a success in all possible worlds in which John comes to the party and which are maximally similar to the actual world w , the party may not be a success in all possible worlds in which John comes to the party late and which are maximally similar to w .

Even if we assume that the antecedent clause of a conditional is non-monotonic, we can compare strengths between readings (ii) and (iii), and between (v) and (vi). This is because the presupposition P_ϕ is always in the consequent clause regardless of whether it is incorporated in the antecedent clause. Consider the following sentence:

- (31) If John gets married, his children will be happy.

For simplicity, we exclude the reading in which the presupposition is projected. Then the two readings to compare are roughly as follows:

4 Veltman (1986) claims that the antecedent clause of a conditional is monotonic. If we assume this together with (15v), the entailment relations shown in (28) hold. But he also claims that the negation of a conditional sentence $(\phi \rightarrow \psi)$ is equivalent to $\circ (\phi \wedge \neg \psi)$. This implies that the conditional has the covert necessity operator. If the necessity operator is regarded as a test operator, just like the possibility operator, then readings (ii) and (iii), and also (v) and (vi) give rise to the same information state as far as they are compatible with the context.

- (32) (i) If ϕ , P_ϕ and ψ .
 (ii) If ϕ and P_ψ , P_ϕ and ψ .
 ϕ : John gets married.
 P_ϕ : John has children.
 ψ : John's children are happy.

If John has children who are happy in all the possible worlds in which John gets married and which are maximally similar to the actual world w , then the same set of possible worlds are those in which John gets married and has children and which are maximally similar to w . And in those worlds, the children will be happy. Thus (32i) entails (32ii). But the other way of entailment does not hold. Suppose that in all the possible worlds in which John gets married and has children and which are maximally similar to w , the children are happy. Those possible worlds may be different from those in which John gets married and which are maximally similar to w . That is, there may be possible worlds in which John gets married and has no children and which are maximally close to w . Then in these worlds, it is impossible that John's children are happy since he has no children. Similarly in (27), readings (ii) and (v) entail (iii) and (vi) respectively, but not vice versa. Then the entailment relations shown in (28) could be maintained.

Among the six readings in (27), however, (iii) and (vi), in which the presupposition triggered in the consequent clause is incorporated into the antecedent clause, are excluded for some reason. I give two examples as evidence for this: one is a conditional sentence and the other a sentence with a universal quantifier. Let's look at conditional sentences first.

- (33) a. If Central Africa becomes a monarchy, its king/(?)queen will represent the country.
 b. If Central Africa becomes a monarchy, it has a king/(?)queen and he/she will represent the country.
 c. If Central Africa becomes a monarchy and it has a king/queen, he/she will represent the country.

In the first sentence, the use of the word *queen* may be surprising when normally a monarchy is expected to have a king. This effect can be

captured only in the paraphrase in which the relevant presupposition is incorporated in the consequent of the conditional, as shown in (33b). If it is incorporated in the antecedent clause, the surprising effect disappears, as shown in (33c). Given that the original sentence sounds odd, we have to say that the presupposition triggered in the consequent clause should not be incorporated in the antecedent clause. Of course, if (33b) is true, then (33c) is true. But this entailment relation cannot be exploited in accounting for why one reading is not available.

Let's consider another example:

- (34) John wants to have one son and one daughter. If he gets married, he will call his son Eugene.
- (35) ?John wants to have one child. If he gets married, he will call his son Eugene.
- (36) ??John doesn't want to have a son. If he gets married, he will call his son Eugene.

In the first example, if John gets married, he is likely to have a son. Thus the conditional is understood as saying that if John gets married, he will have a son and call him Eugene. In the second example, on the other hand, the first sentence implies that if John gets married, he is likely to have one child, but it is neutral as to whether the child may be a son or daughter. In this context the reading that 'if John gets married, he will have a son and call him Eugene' does not seem quite natural. Then, one possibility left is to understand the conditional as 'if John gets married and has a son, he will call him Eugene.' However, the oddity of (35) implies that we cannot get this reading. In the third example, the reading in which the presupposition is incorporated into the consequent clause is overtly avoided. Yet the presupposition is not allowed to go into the antecedent clause. The different acceptabilities of the three examples depend only on whether the presupposition can be incorporated into the consequent clause, that is, whether it is natural to accept that if John gets married, he will have a son in the context. If the presupposition could go into the antecedent clause, there would be no explanation for why (35–36) are awkward.⁵

5 There are examples in which a presupposition in the consequent clause is seemingly incorporated into the antecedent:

Beaver (1995) pointed out that a presupposition from the nuclear scope is not incorporated into the restrictor of the universal quantification.

- (37) a. Every girl brings her boyfriend to the party.
 b. ??She has a boyfriend, and every girl brings her boyfriend to the party.
 c. Every girl who has a boyfriend brings him to the party.
 d. Every girl has a boyfriend and brings him to the party.

A universal quantifier introduces a restrictive clause and a nuclear scope, which correspond to the antecedent and the consequent clause of a conditional. The presupposition from *her boyfriend* is not projected over the universal quantifier, as shown in (37b), since the variable introduced by the pronoun *her* would not be bound. It can only be projected to the restrictive clause, as in (37c), or to the nuclear scope, as in (37d). But (37c) has a different truth condition from (37a). Only when the presupposition is incorporated in the nuclear scope can we get the reading which has the same truth condition as (37a). (37d) entails (37c), but this does not account for why (37c) does not count as a reading of sentence (37a). Moreover, when we consider other quantifiers which have different properties in monotonicity, the entailment relation may go the other way around. Whatever the quantifier may be, no presupposition can be incorporated into the restrictor and change the domain of quantification.

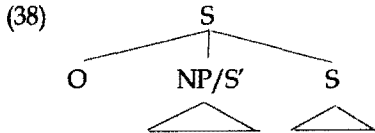
What is common between conditionals and quantifiers is that they have domains over which the operators quantify, and that presuppositions triggered in the consequent clause or in the nuclear scope do not change the domains by being incorporated into the antecedent clause or in the restrictor.⁶ It is not clear to me whether there are semantic

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- (i) If a cat is healthy, it lands on its feet.
 (ii) ??If a cat is healthy, it touches ground and lands on its feet.
 (iii) If a cat is healthy and touches ground, it lands on its feet.

The paraphrase (ii) seems odd even though the original sentence is fine. One reason I can think of is that the consequent clause is a generic sentence, and that there is an implicit restrictor which is provided by the presupposition: the consequent clause roughly means that if it touches ground, it lands on its feet. Since 'if ϕ , then if ψ , χ ' is equivalent to 'if ϕ and ψ , then χ ' we get the paraphrase (iii), instead of (ii).

6 In this respect, the anaphoric analysis of presupposition projection cannot avoid criticism,

reasons for this. One possible explanation is to assume, following Heim (1982), that in natural language these two constructions have the following syntactic structure.



In this structure, when the operator *O* is *every*, the following constituent is NP whose determiner is missing. The NP is followed by a clause to which it is adjoined. When the (often implicit) operator is one for a conditional, the following constituent is *S'*, which begins with something like *if*. It is followed by the main clause *S*. If the structure is interpreted compositionally, presuppositions from the lower *S* are not incorporated into NP or *S'*. This allows us to account for why presuppositions from the consequent clause or the nuclear scope should not be incorporated into the antecedent clause or the restrictor.

So far we have observed various cases, but we have not seen any case where a presupposition has intermediate scope. This might suggest that a presupposition is maximally projected or stays in its place. To argue that presuppositions have various scopes, I need to give evidence that presuppositions have intermediate scope. Consider the following sentence and the schematic representations of its readings:

- (39) If John gets married, he will not let his children inherit his business.
- i. If ϕ , P_ϕ and $\neg(P_\phi \wedge \phi)$
 - ii. If ϕ , $\neg(P_\phi \wedge \phi)$

Structurally the presupposition is triggered within the scope of the negation in the consequent clause, but in the first reading, it is incorporated over the negation operator in the consequent clause. It roughly means that if John gets married, he will have children, and he

either. If presuppositions were anaphoric material and were to be bound by accommodation in a representation like DRSs, as van der Sandt (1992) claims, there would be no reason why they are not incorporated in the antecedent clause or in the restrictor.

will not let them inherit his business. The other reading is the one in which the presupposition goes within the scope of the negation operator, i.e., the reading that if John gets married he may or may not have children, but in either case, no one will inherit his business. But this reading is less likely. This result can be predicted from the property of conditionals that the consequent clause is a monotone-increasing domain.

- (40) a. If John has children, they will succeed his business.
 b. If John has children, they will happily succeed his business.

Sentence (40b) entails (40a). Similarly (39i) is stronger than (39ii). The presupposition in (39) does not seem to project to the main context because the context implies that he will have children after he gets married, but considering this maximally projected reading together with the other readings, the preferred reading is the one in which the presupposition has intermediate scope, and this reading becomes salient when the maximally projected reading is contextually canceled.

If our analysis is right, a presupposition triggered in the antecedent clause of a conditional or in the restrictor of a universal quantifier may, or may not, project, depending on the strengths of the derived readings. Consider the following examples:

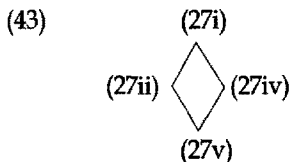
- (41) I don't know whether John has a car, but if he brings his car he doesn't have to walk to the park
 (42) I don't know whether John has a car, but if he does not bring his car, he has to walk to the park.

In these examples, the triggered presupposition does not project to the main context, but stays within the antecedent clause. In (41), the conditional is construed as 'if he has a car and brings it, he won't have to walk to the park.' In (42), the negation operator is added to the antecedent clause. The presupposition can be projected over the negation operator or stays within it. Thus we can get two possible readings: (i) if John has a car and does not bring it, he has to walk to the park, and (ii) if John brings no car regardless of whether he has a car, he has to walk to the park. The latter reading is stronger than the former.⁷

⁷ We cannot claim this if we assume that the antecedent clause of a conditional is not

And it seems to me that the latter is the natural reading in the context. This seems to show that presupposition projection depends on the strengths of possible readings.⁸

Let's go back to our conditional example (26). From the discussion so far, we get four readings out of the six, and they show the following entailment relations.



The diagram indicates that (27i) is the most preferred reading of sentence (26), and one of the two presuppositions can be canceled.

- (44) I don't know if John has a sister or if the woman you talked about is his sister. If John's sister has grandchildren, his children are grown-ups.
- (45) You must find if John has children. If his sister has grandchildren, his children are grown-ups.

In the example (44), the presupposition triggered in the antecedent is canceled, and in (45) the one triggered in the consequent clause is canceled. When one reading is canceled/suspended in the context, that reading is not the only one that is canceled. In (44), (27i) and (27ii) are canceled at the same time. In (45), (27i) and (27iv) are canceled together too. Therefore the remaining readings also constitute a partial order with

monotonic, as Stalnaker and Lewis claim.

8 When the conditional in (42) is negated again, as in (i) below, the preference order is not reversed.

- (i) I don't know whether John has a car, but it is not the case that if he does not bring his car he has to walk to the park

The reason is that even if a conditional is negated, the antecedent clause is not affected. Only the consequent clause or the relation between the antecedent and consequent clauses is negated.

one particular reading indicated as the preferred reading. The case where the two readings (27ii) and (27iv) are compared never occurs.

3. Derivation of Various Readings

In the preceding section, I have claimed that a sentence with presuppositions yield various readings and that readings get stronger when presuppositions are projected. In this section, I will show how a set of readings is derived. A sentence is assumed to be syntactically disambiguated as an input to semantic interpretation. In this paper, I am assuming that presuppositions generate ambiguity in semantic structure. Thus, a syntactically disambiguated sentence can generate more than one reading if it contains a presupposition. In the component of semantic interpretation, there are two possible ways to derive a set of readings from a syntactically disambiguated structure, i.e., syntactic derivation and semantic interpretation. A syntactic derivation needs an extra syntactic level between a syntactic structure and the semantic interpretation, and at that level the sites at which presuppositions are incorporated are determined. This formulation bears an implication that presupposition projection is a syntactic matter. This has been partially successful, as in van der Sandt (1992). However, there are cases where syntactic analysis is not adequate. So I derive a set of readings in the process of semantic interpretation. This allows us to analyze cases where presupposition projection is semantically or pragmatically restricted.

In deriving a set of readings from a sentence, I use a mechanism similar to quantifier storage by Cooper (1983). I divide the semantic interpretation of an expression into an assertion part and a presupposition part. Presuppositions in a sentence, which have the type of context change potentials, are first collected in the presupposition part and then incorporated into the assertion part at various points of the semantic structure.⁹ This makes it possible for each presupposition to have various possible scopes with respect to all operators in higher positions in the syntactic structure of the sentence.

Suppose that a sentence has the following structure:

9 In this respect, our theory is in line with the Russellian theory of scopal variation of presuppositions. Presuppositions are also part of the assertion and the only difference is that they can have various logical scopes which are not restricted by syntactic principles.

$$(46) \quad [{}_X O_1 [{}_Y \dots P_1 \dots [{}_Z O_2 [{}_W \dots P_2 \dots]]]]$$

Here 'O' represents a logical operator. For each presupposition, there is a corresponding assertion part which is not overtly labeled: A_1 and A_2 in relation to P_1 and P_2 respectively. When this sentence is interpreted from the bottom up compositionally, the semantic representation for each syntactic node is given below.

(47)

W	$\langle A_2, \{P_2\} \rangle$	
Z	$\langle O_2[P_2 \& A_2], \{\} \rangle$	$\langle O_2[A_2], \{P_2\} \rangle$
Y	$\langle A_1 \& O_2[P_2 \& A_2], \{P_1\} \rangle$	$\langle A_1 \& O_2[A_2], \{P_1, P_2\} \rangle$
X	$\langle O_1[P_1 \& A_1 \& O_2[P_2 \& A_2]], \{\} \rangle$	$\langle O_1[P_1 \& P_2 \& A_1 \& O_2[A_2]], \{\} \rangle$
	$\langle O_1[A_1 \& O_2[P_2 \& A_2]], \{P_1\} \rangle$	$\langle O_1[P_1 \& A_1 \& O_2[A_2]], \{P_2\} \rangle$
		$\langle O_1[P_2 \& A_1 \& O_2[A_2]], \{P_1\} \rangle$
		$\langle O_1[A_1 \& O_2[A_2]], \{P_1, P_2\} \rangle$

When the node W is interpreted, the presupposition P_2 is collected in the presupposition part. When the interpretation of W is combined with the operator O_2 , the presupposition is either incorporated within the scope of O_2 or remains in the presupposition part. When Y is interpreted, each of the two readings collects the presupposition P_1 in the presupposition part. When the interpretation of Y is combined with O_1 , a subset of the presupposition part is incorporated in the assertion part. Those which still remain in the presupposition part ultimately have a wide scope over O_1 . This process generates six readings in all, as can be predicted. P_2 has three possible scopes with respect to O_1 and O_2 , and P_1 has two possible scopes with respect to O_1 . Since the meaning of a sentence is derived bottom-up compositionally, no presupposition is incorporated into the assertion part of a syntactically lower structure.

Based on the structured meaning analysis, a model and the interpretation function $\parallel \parallel$ is defined below.

- (48) A model is a quadruple $\langle W, E, G, F \rangle$, where W is a set of possible worlds, E the domain of individuals, from which various domains D_τ of type τ are derived, and G the set of assignment functions from discourse markers D to individuals E , and F the interpretation function that maps constants of each type to elements in its domain.
- (49) the set of all possible information states $S^* = \wp(W \times G)$
- (50) Interpretation rules:
- (i) $\| P(i, \dots, j) \| =$
 $\{ \langle \lambda s \{ wf \in s \mid \langle f(i), \dots, f(j) \rangle \in F(P) \}, \{ \} \rangle \}$
 (If $\{ i, \dots, j \} \not\subseteq \text{Dom}(s)$, then undefined.)
 $\text{Dom}(s) = \{ d \in D \mid \exists w \exists f \exists a \text{ } wf \in s, f(d) = a, a \in E \}$
- (ii) $\| \text{NOT } \phi \| =$
 $\{ \langle \lambda s \{ wf \in s \mid \neg \exists g \geq f : wg \in s, P_1, \dots, P_n, R_0 \}, R_1 - X \rangle \mid$
 $\langle R_0, R_1 \rangle \in \| \phi \|, X = \{ P_1, \dots, P_n \} \subseteq R_1 \}$
 – $g \leq f$ iff for all $d \in \text{Dom}(g), g(d) = f(d)$
 – $\text{Dom}(g) = \{ d \in D \mid \exists a \in E : g(d) = a \}$
- (iii) $\| A/\text{SOME}_i \phi, \phi \| =$
 $\{ \langle \lambda s. \{ wg[i] \mid wg \in s \}, P_1, \dots, P_n, R_0, Q_1, \dots, Q_n, T_0,$
 $(R_1 \cup T_1) - (X \cup Y) \rangle \mid \langle R_0, R_1 \rangle \in \| \phi \|, \langle T_0, T_1 \rangle \in \| \phi \|,$
 $X = \{ P_1, \dots, P_n \} \subseteq R_1, Y = \{ Q_1, \dots, Q_n \} \subseteq T_1 \}$
 – $g[i]$: a new function such that
 $g \leq g[i] \& \text{Dom}(g[i]) = \text{Dom}(g) \cup \{ i \}$
- (iv) $\| \phi \text{ AND } \phi \| =$
 $\{ \langle \lambda s. s, R_0, P_1, \dots, P_n, T_0, (T_1 - X) \cup R_1 \rangle \mid$
 $\langle R_0, R_1 \rangle \in \| \phi \|, \langle T_0, T_1 \rangle \in \| \phi \|, X = \{ P_1, \dots, P_n \} \subseteq T_1 \}$
- (v) $\| \text{MIGHT } \phi \| =$
 $\{ \langle \lambda s. s, R_1 - X \rangle \mid \langle R_0, R_1 \rangle \in \| \phi \|,$
 $X = \{ P_1, \dots, P_n \} \subseteq R_1, s, P_1, \dots, P_n, R_0 \neq \emptyset \}$
- (vi) $\| \text{IF } \phi, \text{ THEN } \phi \| =$
 $\{ \langle \lambda s. \{ wf \in s \mid \forall g \geq f [wg \in s, P_1, \dots, P_n, R_0 \rightarrow$
 $\exists h \geq g : wh \in s, P_1, \dots, P_n, R_0, Q_1, \dots, Q_n, T_0] \} \rangle \}$

- $$\begin{aligned}
 & (R_1 \cup T_1) - (X \cup Y) \rangle | \langle R_0, R_1 \rangle \in \| \phi \|, \langle T_0, T_1 \rangle \in \| \psi \|, \\
 & X = \{P_1, \dots, P_n\} \subseteq R_1, Y = \{Q_1, \dots, Q_n\} \subseteq T_1\} \\
 \text{(vii)} \quad & \| \text{EVERY/ALL}_i \phi, \psi \| = \\
 & \{ \langle \lambda s. \{ wf \in s \mid \{ x \mid \exists g[wf \in \{ wf \ i/x \}]. P_1 \dots P_n. R_0 \}] \subseteq \\
 & \{ x \mid \exists g[wf \in \{ wf \ i/x \}]. P_1 \dots P_n. R_0. Q_1 \dots Q_n. T_0 \} \} \rangle, \\
 & (R_1 \cup T_1) - (X \cup Y) \rangle | \langle R_0, R_1 \rangle \in \| \phi \|, \langle T_0, T_1 \rangle \in \| \psi \|, \\
 & X = \{P_1, \dots, P_n\} \subseteq R_1, Y = \{Q_1, \dots, Q_n\} \subseteq T_1\} \\
 & - \ i/x \text{ is an assignment function such that } \ i(i) = x
 \end{aligned}$$

In these rules $s \in S^*$ is an information state. Rule (i) is a rule for atomic sentences, where no presupposition is triggered. Thus the presupposition part of the interpretation is the empty set. Assuming that all NPs have been quantifier-raised, atomic sentences have only discourse referents as their arguments, which must be already in the domain of the discourse referents in the current information state. When a presupposition is triggered in an atomic sentence, it will be collected in the presupposition part. Rule (ii) is the rule for sentences prefixed with the negation operator. In this rule $R = \langle R_0, R_1 \rangle$ is a reading of a sentence ϕ , and R_0 is the assertion part in the form of a context change potential and R_1 is the presupposition part which contains a set of presuppositions. Each presupposition is represented as a context change potential, too. When the sentence is combined with the negation operator, every subset, say $X = \{P_1, \dots, P_n\}$, of the presupposition part is incorporated within the scope of the negation operator and its complement $(R_1 - X)$ remains in the presupposition part. The incorporation of presuppositions is represented in the rule as follows:

$$(51) \quad s. P_1 \dots P_n. R_0, =_{df} R_0(P_n(\dots(P_2(P_1(s))\dots)).$$

That is, the information state s is updated with P_1 , and the resulting information state is labeled s_1 . s_i is updated with P_{i+1} , resulting in s_{i+1} . Finally the resulting information state s_n is updated with R_0 . Let's call the resulting information state s' . Among the pairs of a possible

world and an assignment function in s , those whose assignment functions can be extended to satisfy ϕ and which can thus be included in s' are excluded from s . Since presuppositions in X are incorporated into the assertion part, the presupposition part becomes $(R_1 - X)$. For every other reading of ϕ , we do the same thing. Therefore for each reading $\langle R_0, R_1 \rangle$ of ϕ , we get $2^{\#(R_1)}$ readings.

Rule (iii) is the one for (singular) indefinites. The domain of the discourse referents in the information state is extended by a new discourse referent, and the resulting information state is updated with the common noun phrase ϕ and the rest of the sentence ψ . Every subset of the two presupposition parts, R_1 and T_1 , is incorporated within the scope of the existential operator.¹⁰ Rule (iv) is the one for the conjunction operator. The rule is formulated asymmetrically because a presupposition triggered in the second conjunct may be bound by some expression in the first conjunct. Strength does not change whether a presupposition is incorporated before or after the conjunction operator. Rule (v) is the one for the possibility operator. In this rule every presupposition in the presupposition part may or may not be incorporated within the scope of the operator and each reading of ϕ expands to the number of the subsets of the presupposition part. The possibility operator is taken to be a test operator. So an information state does not hang with MIGHT ϕ unless ϕ or an incorporated presupposition changes the context to the empty set.

In giving the interpretation rule for a conditional, I basically treat it as a material implication. Part of the presuppositions triggered in the antecedent clause are incorporated in that clause and the others are projected to the main context. Similarly some presuppositions triggered in the consequent clause stay in that clause and the rest are projected to the main context. But they do not go into the antecedent clause. The interpretation rule for the universal quantifier is formulated similarly, but the quantifier quantifies over objects rather than assignment functions. Presuppositions from the restrictor may or may not be

¹⁰ This does not actually increase the number of readings since it does not change the strength of a reading for a presupposition to be incorporated inside or outside the existential quantifier. This is only relevant to the problem of whether an anaphoric discourse marker is bound or not.

projected. Those from the nuclear scope also may stay in the nuclear scope or project to the main context, but not to the restrictor. This is not because the rules are formulated as such. It follows from syntactic structures and compositional interpretation.

4. Projection and Cancellation of Presuppositions

Since a sentence yields a set of readings, we understand contexts as sets of information states, each of which is updated with a set of readings again, and yields as many information states as the readings. Notice, however, that each reading is defined with regard to a single information state for a simpler formulation. To avoid type mismatch, a new update operator ' $\times$ ' is introduced. The ' $\times$ ' operator receives a set of information states $S \subseteq S^*$ and a set of readings, and yields a set of information states by the product of the number of readings and the number of information states.

$$(52) \quad S \times \|\phi\| = \{s.P_1 \dots P_n.R_0 \mid \\ s \in S, \langle R_0, R_1 \rangle \in \|\phi\|, R_1 = \{P_1, \dots, P_n\}\}$$

$\langle R_0, R_1 \rangle$, which is a pairing of an assertion and a set of presuppositions, is a reading of ϕ . $P_1, \dots$, and P_n in R_1 are projected presuppositions and thus, they update each incoming information state first. Then the assertion part R_0 changes the resulting information state.

There is a general condition on utterances: they must be informative and must be consistent. In our analysis, multiple readings generate multiple information states in a context, and some readings are canceled after the sentence is followed by several sentences. So the condition must be imposed on both information states and readings:

$$(53) \quad \text{Let } s \text{ be an information state, and } \langle R_0, R_1 \rangle \text{ a reading of a} \\ \text{sentence } \phi, \text{ where } R_1 = \{P_1, \dots, P_n\}. \text{ Then} \\ \emptyset \subset \{w \in W \mid \exists g : wg \in s.P_1 \dots P_n.R_0\} \subset \{w \in W \mid \exists g : wg \in s\}$$

This condition says that a reading may not change an information state to the absurd state, and that it must reduce the size of the information state. When this condition is not met, the combination of the information state and the reading is removed from the resulting context. Let's see how it works.

- (54) If you did not meet *Mary's husband*_i at the party, *he*_i may have stayed at home. However, it is also possible that she is not married.

The first sentence has two readings with respect to presupposition projection. After the interpretation of the first sentence, we get two groups of information states: one from the reading in which the presupposition is projected and the other from the one in which the presupposition is not projected. If the first sentence is followed by a sentence which is compatible with the presupposition, the first group of information states remain in the context. In (54), the information states in the former group become the empty set when updated with the second sentence. Thus they are eliminated from the context. This has the effect of canceling the projected presupposition that Mary has a husband. When this happens, we do not actually go back to the previous sentence to reinterpret it.

There are cases where two presuppositions are incompatible. This can be observed in the following example:

- (55) It is possible that John stopped smoking. It is also possible that he started smoking.

Each of the two sentences generates two readings. If any one presupposition is projected, the information state becomes the absurd state. Thus, neither of the presuppositions are projected. There is a case where no information state in *S* is compatible with any reading of a sentence. This is observed in (4), which is repeated here:

- (4) ??John regrets leaving home. Actually he has not left home.

The first sentence has only one reading in which John has left. The unique reading is incompatible with the second sentence. The reason why the discourse is odd is that there remains no other reading compatible with

the second sentence. I have shown that the so-called presupposition cancelation is just a cancelation of one preferred reading or one preferred information state.

5. Disjunction and Presupposition Projection

Karttunen and Peters (1979) tried to give rules for deriving presuppositions of a sentence compositionally from presuppositions triggered in its constituents. The following is their inheritance rule for the disjunction operator:

$$(56) \quad \text{Pr}(\phi \text{ OR } \psi) = [[\text{Pr}(\phi) \vee \psi] \wedge [\phi \vee \text{Pr}(\psi)]]$$

This rule is motivated by the symmetric nature of disjunctive sentences in presupposition projection. Consider the following sentences:

- (57) a. Either no one at the conference solved the problem, or it was Susan who solved it.
 b. Either it was Susan who solved the problem, or no one at the conference did.

In these examples, the order of the two clauses is different, but in either order, the presupposition in one disjunct that someone solved the problem is not projected due to the other disjunct. Soames (1982) finds a problem with the inheritance condition given in (56). When $\text{Pr}(\phi)$ and $\text{Pr}(\psi)$ are incompatible, (56) entails the following:

$$(58) \quad \phi \vee \psi$$

That is, a disjunctive sentence like (59) is predicted to presuppose the very proposition that it asserts.

- (59) Either Bill has just started smoking or he has just stopped smoking.

The symmetry of presupposition projection in disjunctive sentences also causes problems to the satisfaction theory, as Soames (1989) pointed out. The simplest treatment would be (60).

$$(60) \quad s \parallel \phi \text{ OR } \psi \parallel = s \parallel \phi \parallel \cup s \parallel \psi \parallel$$

However, this rule predicts that all presuppositions in ϕ and ψ would be projected, which is not the case. A more complex rule might be required to be like (61):

$$(61) \quad s \parallel \phi \text{ OR } \psi \parallel = s \parallel \text{NOT } \phi \parallel \parallel \phi \parallel \cup s \parallel \phi \parallel \parallel \psi \parallel$$

This rule is symmetric, and a presupposition in one disjunct can be screened by the other.

However, this rule has two problems. First, exclusiveness between disjuncts is incorporated in the rule, and so the information state always reduces to a new one in which the possibility that ϕ and ψ hold together is eliminated. This causes difficulty in dealing with sentence (62).

(62) Either John solved the problem or Sue did. Maybe both did.

When the information state is updated with the first sentence, all possible worlds in which the two disjuncts hold together are eliminated. When this new information state is updated with the second sentence, it becomes the empty set. Contrary to the prediction, however, the two sentences are not incompatible. Exclusiveness between disjuncts should be a pragmatic implicature, which can be canceled.

The other problem is that there is an asymmetric feature to be captured despite the symmetry in presupposition projection. Compare (60) with sentence (59).

(63) ??Either he_i has just stopped smoking or Bill_i has just started smoking.

When the proper name is not introduced in the previous context or the referent of the pronoun is not independently determined, this sentence is odd since the pronoun is not interpretable. When the interpretation rule is formulated symmetrically, both (59) and (63) would fail to be interpreted. In whichever disjunct the proper name may occur, (the negation of) the other disjunct must be interpreted without the help of the proper name. If there is no previously introduced antecedent, the

interpretation always fails.

In addition to the asymmetry in anaphora, it will be better to maintain the logical relationship between the conjunction and disjunction operators. Due to asymmetric behaviors of presuppositions in conjunctions (and primarily the asymmetry in anaphora), the interpretation rule for conjunctions is formulated asymmetrically. If we are to maintain both the asymmetric behaviors of conjunctions and the logical relation between the conjunction and disjunction operators, the interpretation rule should be like the following:

$$(64) \quad s \parallel \phi \text{ OR } \psi \parallel = s \parallel \text{NOT}(\text{NOT}\phi \text{ AND } \text{NOT}\psi) \parallel \\ = s \parallel \phi \parallel \cup s \parallel \text{NOT}\phi \parallel \parallel \psi \parallel$$

With this asymmetric rule, however, the satisfaction theory has difficulty dealing with sentence (57b): the rule predicts that the presupposition triggered in the first disjunct is always projected.

Our analysis allows us to adopt the interpretation rule in (64), which captures the asymmetry in anaphora and maintain the logical relation between the conjunction and disjunction operators.¹¹ With this rule, a presupposition triggered in the second disjunct can be canceled by the first disjunct, just as with (61). Consider the case where a presupposition is triggered in the first disjunct. If this presupposition is projected to the main context, the input information state for the interpretation of the second disjunct will contain the projected presupposition. When the presupposition is incompatible with the assertion ψ , the updated information state becomes the empty set. If the second disjunct contains the same information as the presupposition, this will make the disjunct uninformative. Thus, we get only the reading in which the presupposition of the first disjunct is not projected. There is no way of a presupposition from the second disjunct going into the first disjunct because compositional derivation of readings prevents it.

Our theory predicts that a presupposition in any disjunct is preferred to be projected. When it is projected, the information state only has the possible worlds in which the presupposition holds. If the presupposition is not projected but stays in one disjunct, the disjunct may reduce the

11 This does not mean that we need the interpretation rule in (64). Insofar as presupposition projection is concerned, our theory can adopt the rule in (60) too.

local context to the set of possible worlds in which the presupposition holds, but the other disjunct still allows for the possible worlds in which the presupposition does not hold. Thus, the resulting information state still includes possible worlds in which the presupposition does not hold. This shows that the projected reading leads to a stronger reading.

Note that the paraphrase of a disjunctive sentence with the negation and conjunction operators in (64) was only used to show the logical relation between the conjunction and disjunction operators. If we actually use the paraphrase in interpreting a disjunctive sentence, we will get more readings than we want because more operators generate more readings. To avoid this, we only adopt the basic idea of (64) in formulating the interpretation rule for the disjunction operator.

$$\begin{aligned}
 (65) \quad \| \phi \text{ OR } \psi \| = \{ & \langle \lambda s(s.P_1 \dots P_n.R_0 \cup \{wf \in s | \\
 & \neg \exists g \geq f: wg \in s.P_1 \dots P_n.R_0\}.Q_1 \dots Q_n.T_0), \\
 & (R_1 \cup T_1) - (X \cup Y) \rangle | \langle R_0, R_1 \rangle \in \| \phi \|, \langle T_0, T_1 \rangle \in \| \psi \|, \\
 & X = \{P_1, \dots, P_n\} \subseteq R_1, Y = \{Q_1, \dots, Q_n\} \subseteq T_1 \}
 \end{aligned}$$

Then the logical relation is only implicitly captured in the interpretation rule. We actually did the same thing in the interpretation rule for conditionals in (50vi). They could be paraphrased with the negation and conjunction operators, but it would generate too many readings.

6. Conclusion

In our theory a set of readings is derived from a sentence and if the strongest reading is canceled or suspended by the context, the next strongest is adopted as the interpretation of what the speaker has intended. Unlike the satisfaction theory, a presupposition is canceled globally, not by the local context. In this respect, I am following Gazdar (1979). One drawback of Gazdar's theory was that his theory could not deal with the sub-sentential level. In his theory, a triggered presupposition is the one which is checked with the context. In my theory, a triggered presupposition is construed with various strengths depending on the structure of the sentence and each construal is checked with the context. One very good point of Gazdar's strategy is that it is a "global

cancellation theory," (in Beaver's (1997) terms). Even when a presupposition is canceled by some information in a later utterance, it can be explained without difficulty. Our theory can deal with presupposition cancellation at sub-sentential level and at the level of texts and thus, can be regarded as a more refined version of Gazdar's theory. And when a presupposition is canceled after a few sentences, our theory needs no backtracking to correct the original interpretation, but just removes a set of information states in which the presupposition holds.

What could be the motivation of this bias towards stronger readings? Consider what would happen if the opposite were the case. If the weakest one were preferred, then there would be no reason for searching for a stronger reading. First, if the weakest reading were consistent with the context, then it would be the reading we are trying to find. So we would not need to find a stronger reading. If the weakest reading were not consistent with the context, then it would be no use to find a stronger reading because if a weaker reading did not hold in a context, neither would a stronger one. Then the linguistic system does not need to provide a set of readings. And the distinction between different readings would remain vague in the actual use of language. That is, when a weaker reading is consistent with the context, it is not distinctively recognized whether or not a stronger reading holds in the context.

An intermediate reading cannot be a default reading because when there is more than one intermediate reading, we cannot decide which reading is intended by the speaker. To disambiguate which reading the speaker has intended, we would always need an additional expression. Therefore the natural choice is to take the strongest reading as a default reading. If the reading is not consistent with the context, the next strongest reading is taken to be what the speaker intended. It will always be clear to both the speaker and the audience which reading is being considered. And a set of readings provided by the linguistic system is put to full use in actual uses of language. This also enhances economy in using language in that we can convey more informative meanings with simpler expressions than when we exploit the weakest readings.

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Received: 2000. 4. 20

Accepted: 2000. 12. 13